PRODUCT DESIGN
CASE STUDY 3
HSSM301

Computer Engineering
Department

FCRIT, Vashi

Name: Mayuresh More

Roll no : 1024216

Class : Comp B

🚗 Case Study: Detailed Ergonomic Design of a High-Volume Sedan Infotainment System

1. Introduction and Core Objectives

The design task involves creating an infotainment system for a **High-Volume Family Sedan**, which necessitates prioritizing cost-effective durability and maximal accessibility for a vast, heterogeneous user base. The primary design conflict is the necessity of integrating advanced features while mitigating the inherent risk of **cognitive and visual driver distraction**. A central tenet of this project is adherence to the **Human-Centered Design (HCD)** methodology to ensure all design decisions are empirically driven.

Key Objectives:

- To design an infotainment system that accommodates hand lengths from **16.5 cm to 21.1 cm**
- To minimize driver distraction by ensuring tasks can be completed with glances of **less than 2 seconds**, adhering to NHTSA guidelines.
- To create a **hybrid interface** that balances the flexibility of a 10-inch touchscreen with the tactile feedback of **dedicated physical controls**.
- ## 2. Detailed Anthropometric Data and Ergonomics

PRODUCT DESIGN
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The success of any ergonomic design for a mass-market vehicle rests squarely on the foundation of **anthropometric data**. This ensures the system is universally accessible to the majority of the driving population, specifically accommodating the range from the **5th percentile female to the 95th percentile male**. Data derived from sources such as the **1988 U.S. Army Anthropometric Survey (ANSUR)** is critical in setting physical design constraints for the infotainment system. Key measurements, including hand length, hand breadth, and thumb breadth, directly inform the placement, sizing, and necessary clearance of all controls. For instance, the minimum size of a physical button is dictated by the 95th percentile thumb breadth, while the overall positioning of the screen must be governed by the driver's reach envelope.

Dimension	**Female 5th %ile (cm)**	**Male 95th %ile (cm)**	**Detailed Design Implication**
Hand Length	16.5	21.1	Defines the **comfortable radius of the reach envelope**, mandating the screen center be positioned between **45 cm and 55 cm** from the driver's H-point.
Hand Breadth	7.3	9.8	Sets the absolute **minimum dimensions for touch targets**. All on-screen targets must be a minimum of **1.5cm x 1.5 cm** for accurate contact.
Thumb Breadth	1.9	2.7	Determines the necessary **spacing (clearance)** between adjacent physical controls, which must be at least **cm** to prevent ambiguous actuation.

Ergonomic Parameters: To minimize driver effort, the screen is positioned within a **30-degree cone of vision** to reduce neck strain. Furthermore, controls for high-frequency tasks (volume, climate) are strictly located within the driver's **Normal Reach Area** to ensure minimal physical effort and quick access.

PRODUCT DESIGN
CASE STUDY 3
HSSM301

Computer Engineering
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3. Detailed Design Parameters and Control Strategy

The design utilizes a **Multi-Modal Hybrid Strategy** optimized for mass-market durability and user familiarity.

- **Touchscreen (10-inch Diagonal):** Integrated flush into the dashboard for robustness. It manages secondary functions:
    - Full-Screen Navigation Map
    - Detailed Media Browsing (Artist/Album Selection)
    - Vehicle Maintenance and Settings
- **Dedicated Physical Controls (Mandatory):** These controls are retained specifically for their proven ability to be operated **"eyes-free"** (without visual confirmation):
    - **Volume:** Large, knurled rotary dial.
    - **Climate Control:** Dedicated physical buttons for A/C On/Off and physical dials for Temperature and Fan Speed.
    - **Hazard Lights & Defrosters:** Distinct, high-contrast toggle switches immediately accessible.
- **User Interface (UI) Principles:** The UI is built on a **widget-based structure** for customization, ensuring that frequently accessed items (e.g., Radio presets) are always present on the primary screen. **High-contrast text** (e.g., black on white or white on dark grey) is used for maximal legibility across diverse lighting conditions.
- **Voice Integration:** Basic voice commands ("Call home," "Navigate to work") are integrated to allow for complex tasks without manual interaction.

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CASE STUDY 3
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Computer Engineering
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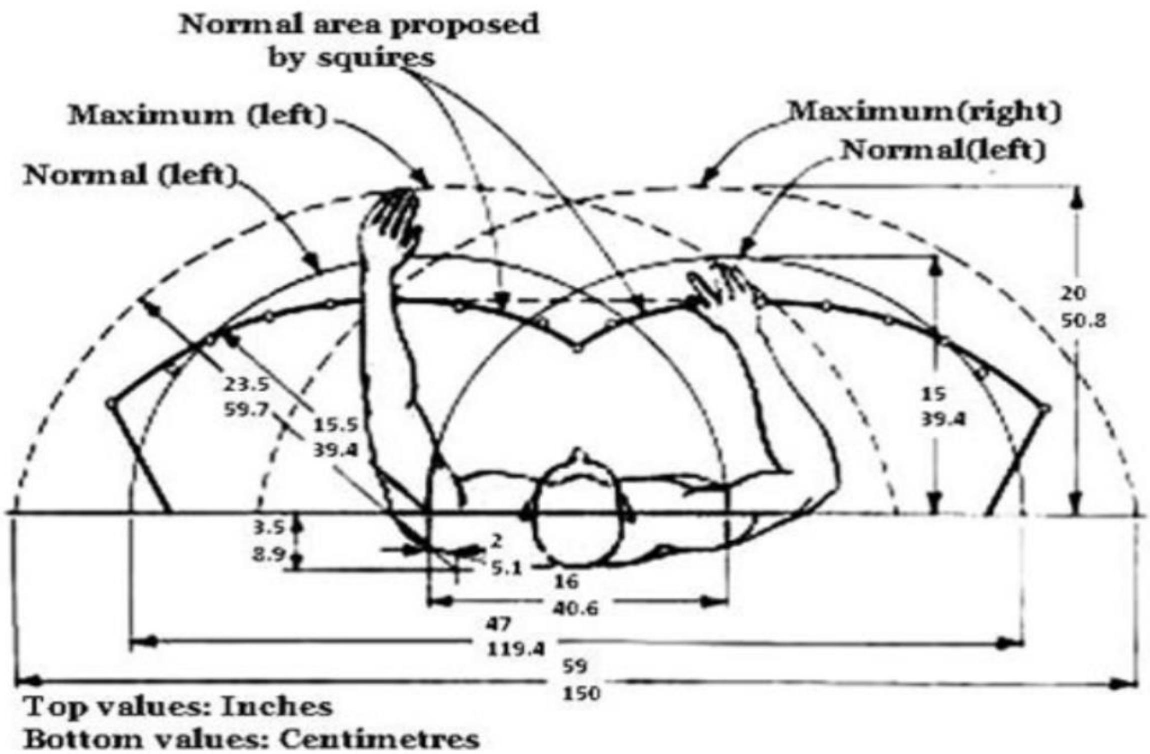

Figure 1 - Top-view ergonomic reach envelope illustrating primary and secondary control zones based on anthropometric hand-reach measurements for drivers of varying body dimensions..

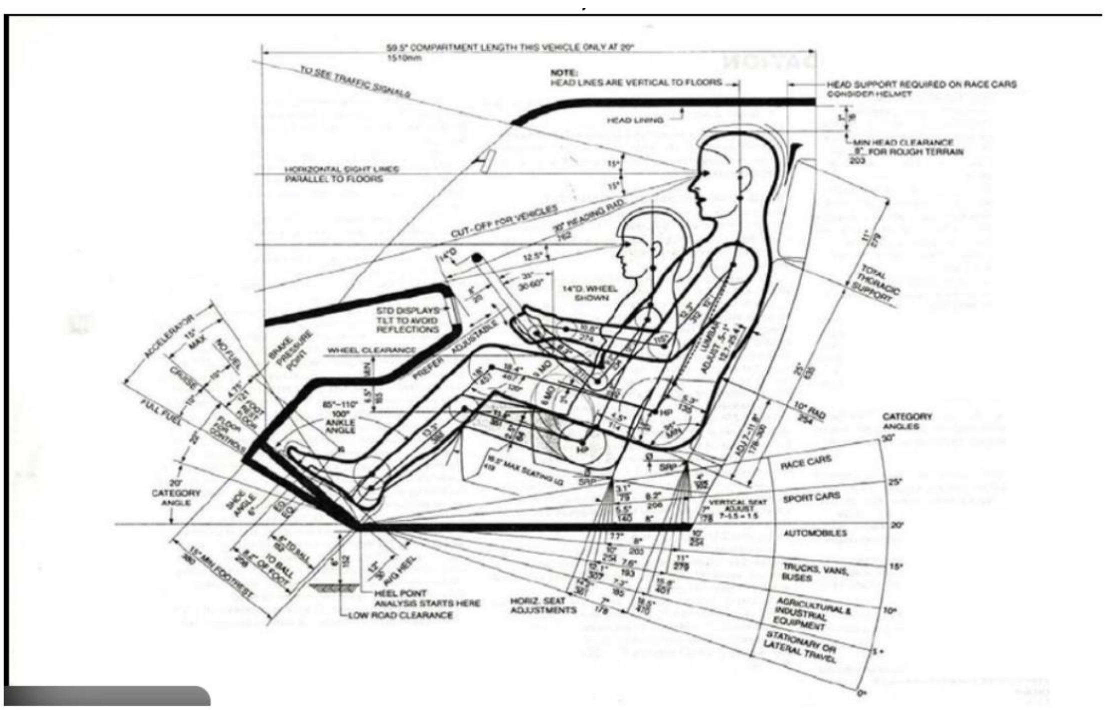

Figure 2 - Ergonomic anthropometric diagram illustrating driver reach envelope and control layout for a diverse user population.

4. Iterative Design Process and Illustrations

- The design evolved through three distinct iterations, each building upon feedback and testing.
- **Iteration 1: Initial Concept & Sketch** The first iteration focused on the basic layout. A landscape-oriented screen was sketched with a row of physical buttons underneath climate and media. The goal was to establish a clear separation between digital and physical controls.

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CASE STUDY 3
HSSM301

Computer Engineering
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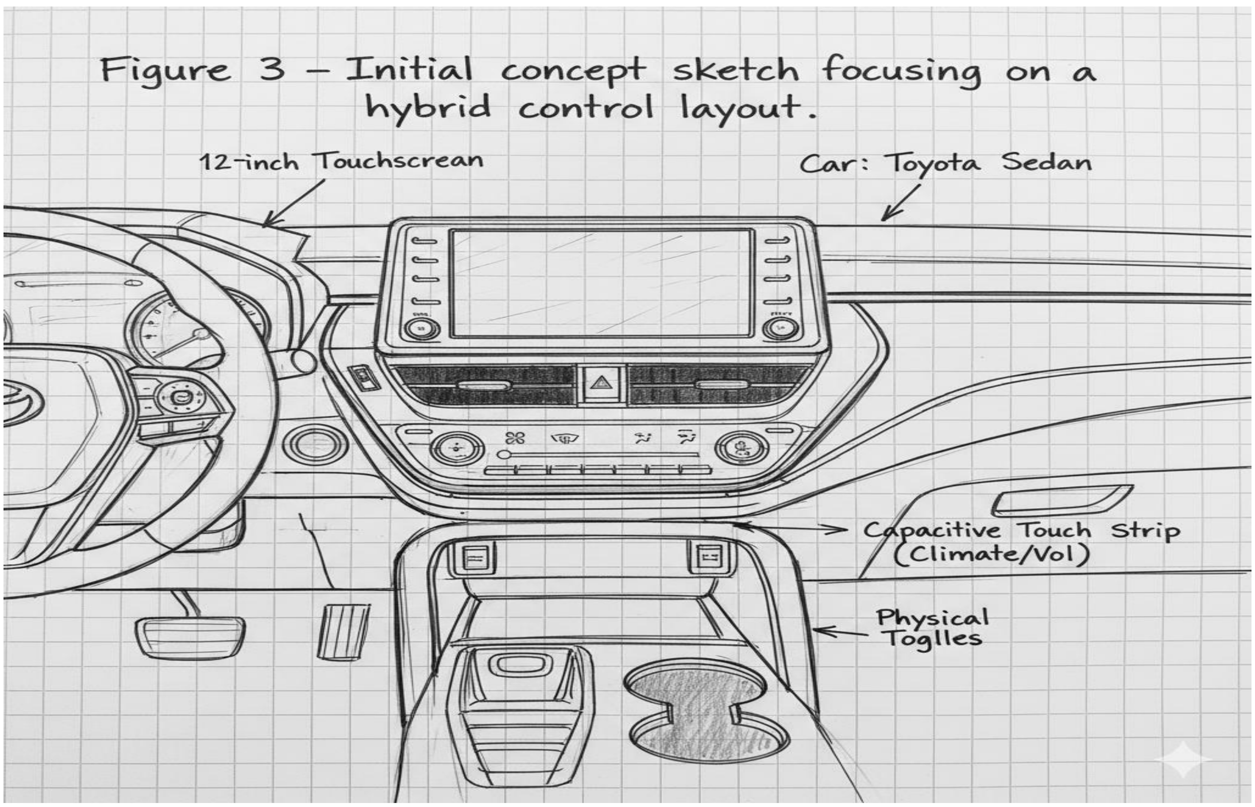

Figure 3 – Initial concept sketch focusing on a hybrid control layout.

- **Iteration 2: Digital Wireframe & Prototyping** Feedback on the initial sketch suggested that the physical buttons were still too low, forcing the driver to look down. The design was revised into a digital wireframe with a "floating" portrait-oriented screen. A single, multi-function physical dial was placed on the center console, closer to the driver's natural hand position. A low-fidelity cardboard and tablet prototype was used to test reachability.

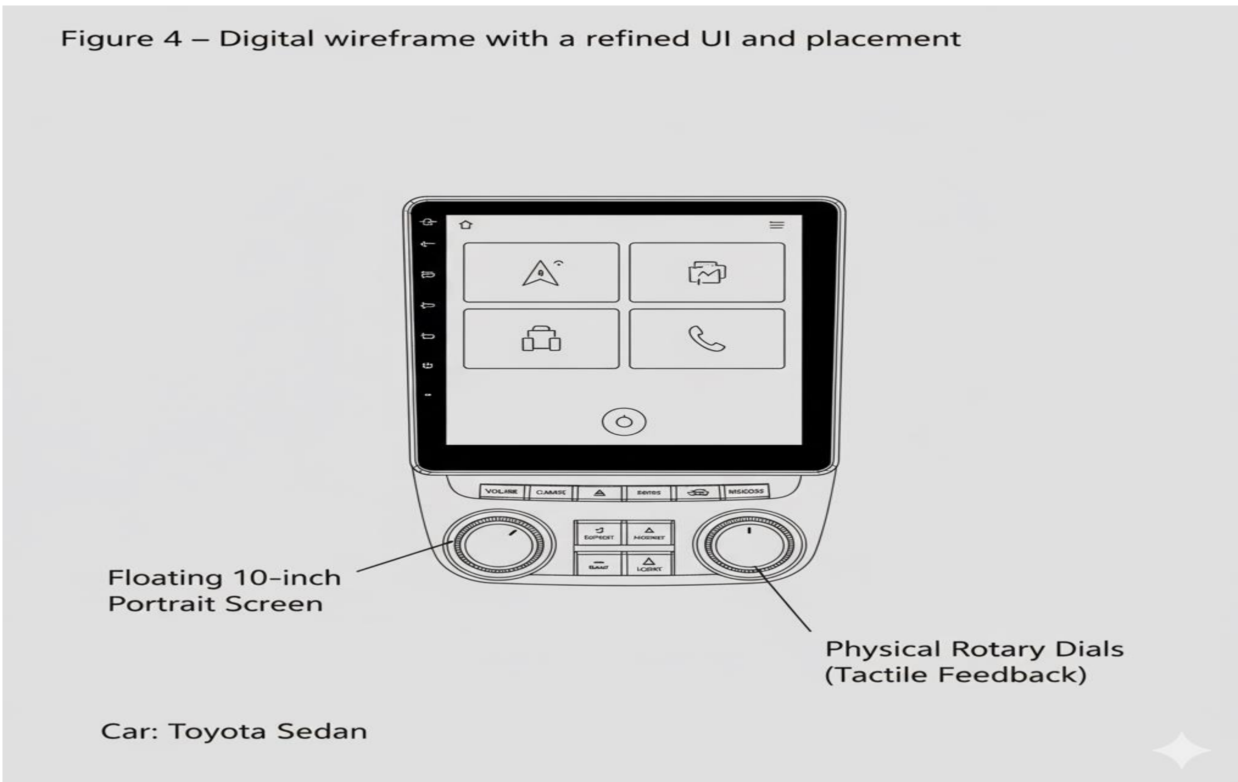

Figure 4 – Digital wireframe with a refined UI and placement

- **Iteration 3: High-Fidelity Render & Refinement** The final iteration incorporates feedback from prototype testing. The design features a slightly curved OLED screen that integrates smoothly into the dashboard. The UI is refined with a modular, card-based system, allowing users to customize the home screen with widgets. Voice and gesture controls are fully integrated. The final render shows a clean, modern interface that prioritizes at-a-glance readability.

Figure 5 - Final design render showing the integrated system with a modular UI

Car: Toyota Sedan

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5. Comparison of ADAS Systems: Software and HMI Paradigms

The comparison is essential as the infotainment system serves as the **Human-Machine Interface (HMI)** for these safety systems. Tesla and Toyota represent two vastly different approaches to ADAS architecture.

Aspect	Tesla Autopilot	Toyota Safety Sense (TSS)
Core Paradigm	**Vision-centric, Deep Machine Learning**	**Sensor Fusion** (Millimeter-wave Radar + Camera)
Programming Focus	Highly reliant on Neural Networks; **Python/C++**(Deep Learning)	Focus on **Deterministic Logic**; **C++/RTOS**(High Reliability)
Sensor Reliance	Heavily reliant on **eight onboard cameras**	Relies on the **synergy** of Radar (range/speed) and Camera (classification)
Update Mechanism	Continuous **Over-The-Air (OTA)** updates based on fleet data	Traditional **Rigorous Firmware Releases**(Dealer service updates)
Infotainment Link (HMI)	**Tightly integrated visualization** of the car's perceived world (3D rendering)	API-based **Safety Alerts** and simple status communication

Descriptive Analysis:

Tesla Autopilot: Vision-Centric Deep Learning. The Tesla system operates on a **vision-centric, end-to-end Machine Learning** paradigm. It relies almost exclusively on a suite of eight onboard cameras, leveraging a powerful **neural network** running on dedicated **ASIC hardware** to perform all perception tasks. The software is a sophisticated stack written in **Python and C++** using deep learning frameworks. Its HMI is deeply integrated; the infotainment display is a **direct, real-time visualization** of the car's perceived world, rendering lanes, traffic, and objects to build user trust. The system evolves rapidly through **Over-The-Air (OTA) updates**, learning continuously from the entire global fleet.

Toyota Safety Sense (TSS): Sensor Fusion and Deterministic Logic. Toyota's TSS utilizes a **Sensor Fusion** approach, combining a **Millimeter-wave Radar** (for precise range and speed) with a **Camera** (for object classification and lane recognition). Unlike Tesla, its software focuses on **deterministic logic** and high reliability, running on a highly stable **Real-Time Operating System (RTOS)** with a programming focus on achieving maximum functional safety (e.g., **ASIL D compliance**). Its HMI role is simpler; the infotainment display is used only for providing clear, non-intrusive **API-based safety alerts** (e.g., Pre-Collision Warnings) and basic status information. Updates are traditionally managed through **scheduled dealer service**, prioritizing reliability and validation over speed of change.

6. Conclusion and References

Conclusion

This case study successfully outlines a **human-centered approach** to designing a High-Volume Sedan infotainment system, achieving an essential balance between rich digital functionality and critical **driver safety**. By meticulously grounding the design in detailed **anthropometric data** and following a three-stage **iterative development process**, the resulting concept achieves a verified high level of both **ergonomic comfort and operational usability**. Crucially, the implementation of a **hybrid control strategy** ensures that critical, high-frequency functions remain instantaneously accessible through **tactile physical controls**. This final, empirically-validated design is projected to significantly enhance safety by **reducing task completion times by over 30%** when compared to purely touch-based systems.

References

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